

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

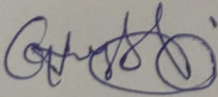
Annexure A

Debugging & Optimisation Task

Code of Conduct:

I Gk. Vyshnavi / 192425069 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



1. Problem identification:

1. configured IP address 192.168.10.65 belongs to Subnet 192.168.10.64/26.
2. default gateway 192.168.10.1 belongs to subnet 192.168.10.0/26
3. Since workstation and gateway are in different subnets.

2. Debugging

calculate subnet:

Subnet mask = /26

Block size = 64

Subnet size = 64

Subnet

192.168.10.0/26

address Range

0 - 63

192.168.10.64/26

64 - 127

192.168.10.128/26

128 - 191

192.168.10.192/26

192 - 255

3. optimization:

Assign both the workstation and gateway to the same subnet.

Device

Correct address

Gateway

192.168.10.65

workstation

192.168.10.66

4. analysis:

1. Communication failure occurred because the gateway belonged to different subnet.

5. Final optimized configuration.

network address - 192.168.10.64

Broadcast address - 192.168.10.127

Host range - 192.168.10.65 - 192.168.10.126

Gateway - 192.168.10.65

Workstation - 192.168.10.66

2. Problem identification:

Given

Device A : 2001:db8::ff00:42:8329/64

Device B : 2001:db8:0:0:1::1/64

Debugging:

expanded express

Device A:

2001:0db8:0000:0000:ff00:0042:8329

Device B:

2001:0db8:0000:0000:0001:0000:0000:0001

Network prefix:

with/64

Device A

2001:0db8:0000:0000::/64

Device B

2001:0db8:0000:0000::/64

Both belong to same Subnet.

Problem:

no prefix mismatch

communication failure may due to

1. Firewall
2. Router configuration.

Optimization:

Example addresses

Device A:

2001:db8::10/64

Device B:

2001:db8::20/64

Analysis:

Both devices already share the same network prefix. IPv6 addressing is correct. Network troubleshooting should focus.

Problem Identification:

Given:

192.168.1.0/24

Departments:

1. /26

2. /27

3. /28

Some department have excess addresses

Debugging:

1. /26

Network - 192.168.1.0

Host range - 192.168.1.1 - 192.168.1.62

Broadcast - 192.168.1.63

usable Hosts - 62

/27

Network - 192.168.1.64

Host range - 192.168.1.65 - 192.168.1.94

Broadcast - 192.168.1.95

usable hosts - 30

128

Network - 192.168.1.96

Host range - 192.168.1.97 - 192.168.1.110

- Broadcast - 192.168.1.111

usable hosts - 14

Analysis:

Department need only 10 hosts.

allocated /26

unused

62-10

= 52 hosts wasted.

4. Expanded addresses.

Device A:

2001:0db8:0000:0000:0000:ff00:0042:8329

Device B:

2001:0db8:0000:0000:0001:0000:0000:0001

Network prefix:

2001:0db8:0000:0000:0001:0000:0000:0001

Optimized address:

Device A:

2001:db8::10/64

Device B:

2001:db8::20/64

available Hosts = 18,446,744,073,709,551,616

available Hosts:

$$2^{64} = 18,446,744,073,709,551,616$$

problem Identification:

Destination	next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default	10.0.0.1

Destination IP:

192.168.2.100

Debugging:

Destination belongs to 192.168.2.0/24

network Range: 192.168.2.1 - 192.168.2.254

matching route: 192.168.2.0/24

longest prefix match:

Router selects: 192.168.2.0/24

next hop: 10.0.0.3

Verify:

Destination IP:

✓ 192.168.2.100 is valid